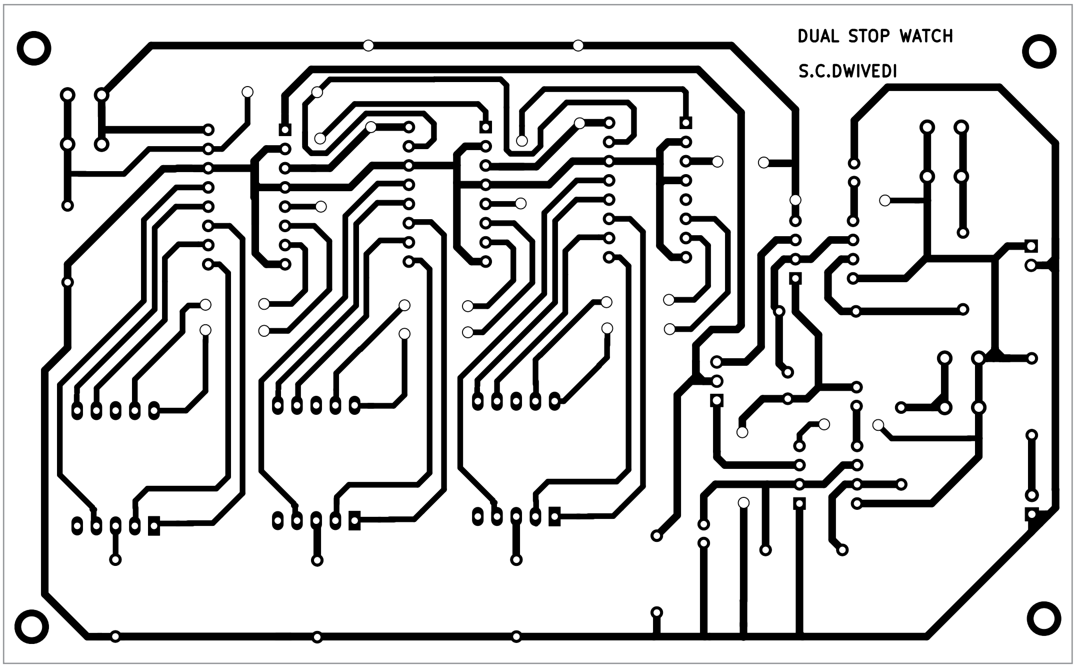

Fig. 3: Actual-size PCB layout

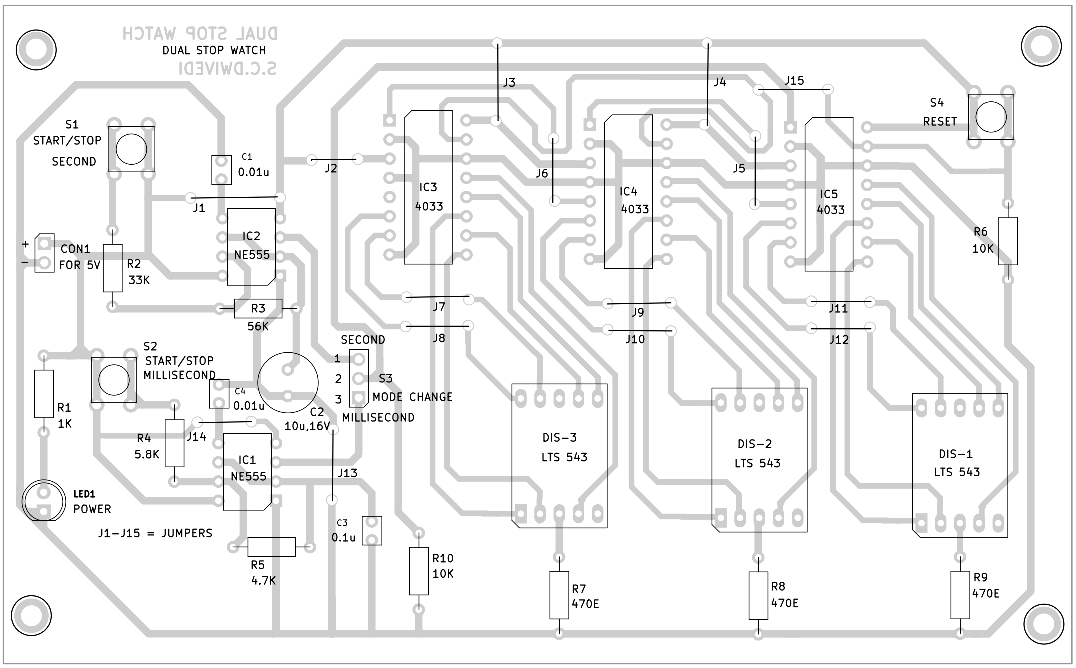

Fig. 4: Component side of the PCB

stopwatch operates by triggering the 7-segment decoder 4033 with pulses from the astable multivibrators.

The working of the stopwatch is straightforward. To illustrate, if you want to measure 50 seconds, set S3 to the seconds position, press reset switch S4, displaying 000. Press start switch S1 until the display shows 050, then release switch S1 to stop counting.

For measuring 500 milliseconds, follow a similar process with S3 in the millisecond position, pressing reset switch S4, displaying 000. Press start switch S2 until the display shows 500, then release S2 to stop counting.

Construction and testing

An actual-size, single-sided PCB layout for the dual stopwatch is shown in Fig. 3, with the component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box, placing LED1 and CON1 on the front side. Position the PCB to allow easy viewing of the three 7-segment displays for time reading. Fix switches S1 through S4 on the front or upper side for convenient access.

LED1 should illuminate upon connecting a 5V adaptor across CON1, indicating the dual stopwatch is ready for use. EFY

Bonus. You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/live-diy-dual-mode-stopwatch*

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